Lecture 3: Diffusion

Many easy steps, and what they cost

Mehmet Kurt, Tianyi Ren
University of Washington

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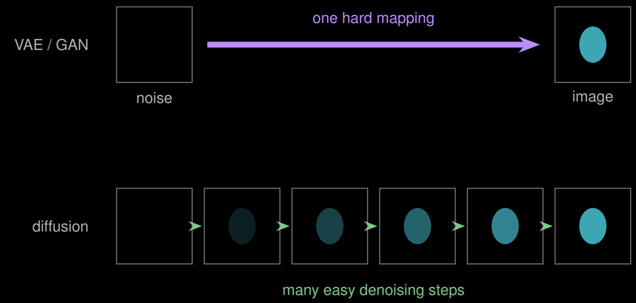
Diffusion: many easy steps

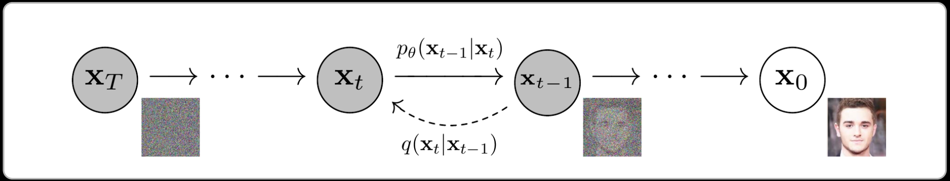
Both models tried to do it in one jump



The idea

Learning "noise $\rightarrow$ anatomy" in one shot is hard, and both previous models paid for it — with blur, or with instability. Learning **"slightly less noisy than this"** is easy, and we can afford to do it a thousand times.

Forward process: destroy the image on purpose



Read it **right to left**: the dashed $q(x_t \mid x_{t-1})$ adds a little Gaussian noise, T times, until nothing is left.

Jump to any t directly

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$$

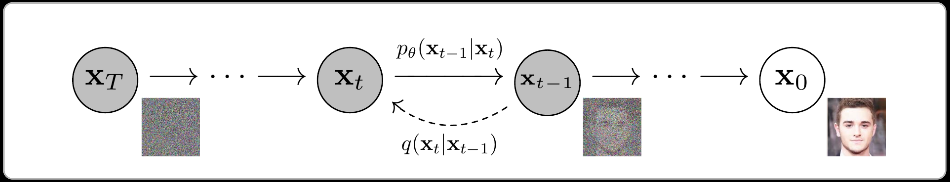
No loop at training time — pick a random t and noise the image in one line.

Nothing is learned

The forward process is **fixed**: a chosen schedule $\beta_1 \dots \beta_T$, no parameters.

Ho, Jain & Abbeel, *Denoising Diffusion Probabilistic Models*, NeurIPS 2020, Fig. 2.

Reverse process: learn to undo one step



Now read it **left to right**: $p_\theta(x_{t-1} \mid x_t)$ is the **learned** half — a network that removes a little noise.

What it sees

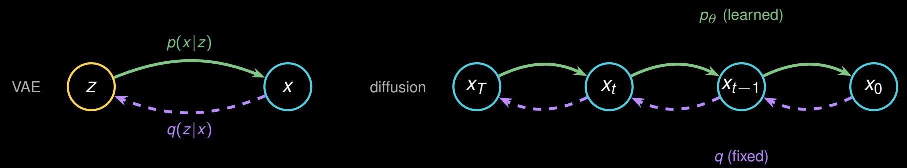
The noisy image x_t *and* the timestep t, so one set of weights covers every noise level. Architecture: a **U-Net** — Module 3, unchanged.

Why it trains

Each step is a small, well-posed denoising problem with a clear target. No adversary, no averaging over the output space.

Same figure, opposite direction: solid arrows p_θ are learned, dashed q is fixed.

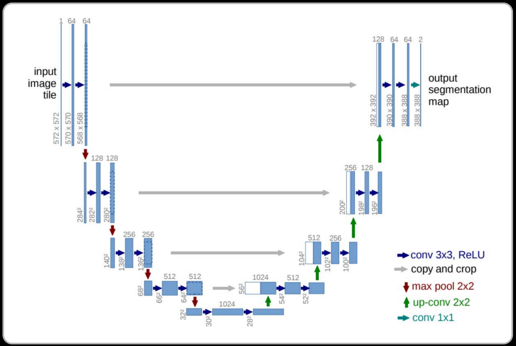
You have seen this before: diffusion is a deep VAE



	VAE	**Diffusion**
Latent size	smaller than x	*same* size as x_0
Encoder q	learned	**fixed** (just added noise)
Steps	one	T, typically 1000

Same encode/decode skeleton. Diffusion simply makes the encoder trivial and the decoder gradual, which is exactly why it trains where the VAE blurs (Lecture 1) and the GAN wobbles (Lecture 2).

What is the denoiser? Your U-Net



Same shape, new job

Input and output are the same size and skip connections preserve detail — exactly what denoising needs. It predicts **noise** instead of a mask, and the timestep t is embedded into every block so one network covers all T noise levels.

Worth saying out loud

The U-Net was invented for **biomedical segmentation** (Module 3), and Stable Diffusion, DALL·E 2 and Imagen all run on it — a rare case of mainstream ML taking its architecture *from* an application domain rather than the reverse.

The training objective is embarrassingly simple

Predict the noise

$$\mathcal{L} = \left\| \epsilon - \epsilon_\theta(x_t,\, t) \right\|^2$$

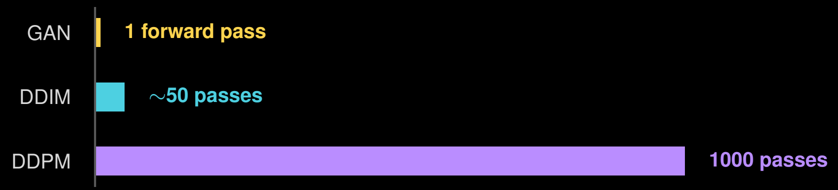
We know ϵ — we added it ourselves. The label is free.

That is the entire loss. No KL, no discriminator, no balancing act.

```
for x0 in loader:
  t = randint(1, T)
  eps = randn_like(x0)
  xt = sqrt(abar[t])*x0 \
      + sqrt(1-abar[t])*eps
  loss = mse(unet(xt, t), eps)
  loss.backward(); opt.step()
```

Ho, Jain & Abbeel, *Denoising Diffusion Probabilistic Models*, NeurIPS 2020.

Sampling, and what it costs



The trade

Diffusion buys stability and sample quality with **inference compute**. A GAN generates in one pass; DDPM needs a thousand.

Two ways to cut it

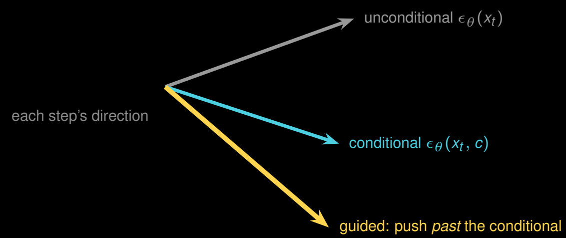
Better schedulers — DDIM, DPM-Solver: same trained model, fewer steps (1000 → 20–50).

Distillation — train a few-step student to imitate the many-step teacher. Changes the model, not just the sampler.

Song, Meng & Ermon, *Denoising Diffusion Implicit Models*, ICLR 2021.

Conditional sampling: steering the denoiser

Unconditional sampling gives you *a* plausible scan. You almost always want **a particular one**.



Classifier-free guidance

$$\tilde{\epsilon} = \epsilon_\theta(x_t) + w\left[\epsilon_\theta(x_t, c) - \epsilon_\theta(x_t)\right]$$

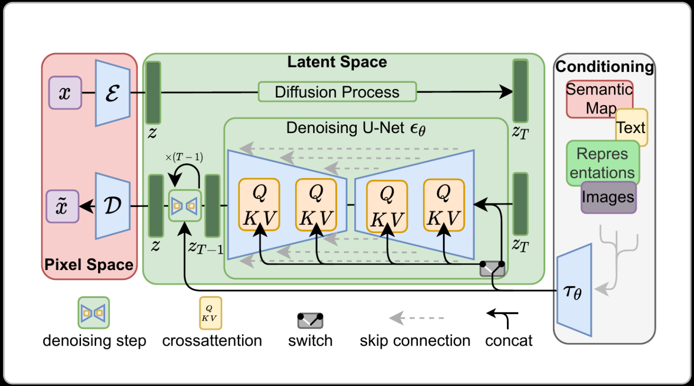
Train one model to do both, then **extrapolate** along the difference.

The dial

w trades **fidelity to the condition** against **diversity**. Turn it up and every sample obeys the prompt — and they all start to look the same.

Ho & Salimans, *Classifier-Free Diffusion Guidance*, 2022. Conditioning c enters the U-Net by cross-attention.

Latent diffusion (a.k.a. Stable Diffusion) — the VAE comes back



Pixel space (left)

$\mathcal{E}$ and $\mathcal{D}$ are a **VAE** — the model from the start of the lecture, now *compressing* rather than generating.

Latent space (middle)

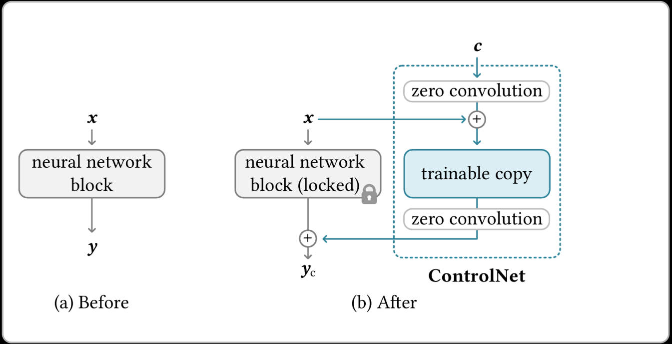
The denoising U-Net runs **here**, not on pixels — typically 8$\times$ smaller per side, so $\sim$**64$\times$ fewer values**.

Conditioning (right)

Text, semantic maps, images — injected by **cross-attention** (Q, KV). What makes diffusion controllable.

Rombach, Blattmann, Lorenz, Esser & Ommer, *High-Resolution Image Synthesis with Latent Diffusion Models*, CVPR 2022, Fig. 3.

ControlNet: conditioning on a mask, not a sentence



How it works

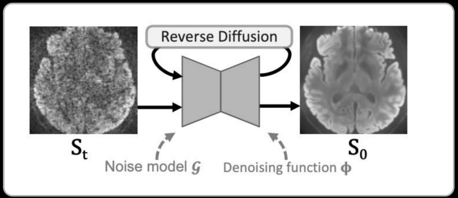
Lock the pretrained diffusion model. Train a **copy** of its blocks on the new condition, joined by **zero convolutions** so the copy starts as a no-op and cannot damage the frozen model.

Why this matters here

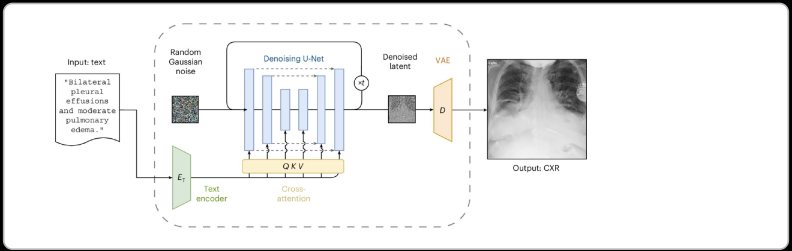
Condition on a **segmentation mask** and you generate an image that matches it — a synthetic scan whose ground truth you already know. Free training pairs for Module 3's problem.

Zhang, Rao & Agrawala, *Adding Conditional Control to Text-to-Image Diffusion Models*, ICCV 2023, Fig. 2.

Diffusion in medical imaging, today



Denoising. Treat a noisy acquisition as x_t and run the reverse process to x_0 — **no clean/noisy pairs needed**, so it trains on the scans you already have.



Synthesis. Stable Diffusion, conditioned on a report: text encoder $\to$ cross-attention $\to$ VAE decoder. Every component is from the last four slides.

Figures via Yeung-Levy & Burgess, BIODS 276 Lecture 7, p. 31. Sources: *Self-Supervised Diffusion MRI Denoising with Generative Diffusion Models*; *A vision–language foundation model for the generation of realistic chest X-ray images*.